Ray diagrams with objective, intermediate and final image planes
Dashed lines: real specimen images · fixed 256 × 256 detector

IL1 -28% · 1.588 A
Specimen-equivalent defocus -10.64 µm

IL1 -20% · 1.765 A
Specimen-equivalent defocus -4.45 µm

IL1 -10% · 1.985 A
Specimen-equivalent defocus -1.45 µm

IL1 +8% · 2.382 A
Specimen-equivalent defocus +0.71 µm

Objective · z from specimen (mm)
Intermediate · z from specimen (mm)
Detector · z from specimen (mm)

Objective
Objective image
188.19 mm
IL1
IL2
IL3
PL1
IL2 image · 311.61 mm
IL2 image · 310.50 mm
IL2 image · 306.94 mm
IL1 image · 254.90 mm
IL2 image · 330.12 mm
IL1: 124.2 mm (virtual)
IL3: 512.7 mm (intercepted by next lens)
IL1: -5.9 mm (virtual)
IL3: 498.0 mm (intercepted by next lens)
IL1: 425.4 mm (intercepted by next lens)
IL3: 467.8 mm (intercepted by next lens)
IL3: 277.8 mm (virtual)
Final image (PL1)
410.221 mm
410.213 mm
410.186 mm
410.311 mm
Detector − image = +308.53 mm
Detector − image = +308.54 mm
Detector − image = +308.56 mm
Detector − image = +308.44 mm
Detector

x (µm)
x (mm)